Foundation Mathematics: Sets

Sample marked worksheet for LearnLens AI's hackathon demonstration

DEMO DATA ONLY Every name and answer is invented. Do not present as real student evidence.

Instructions

Answer each question briefly. The answer samples and teacher comments below are authored only to exercise the software flow.

1. Set-builder notation (2 marks)

Write B = {3, 9, 27, 81} in set-builder form.

Sample response: B = {3, 9, 27, 81}

Illustrative marks: 1 / 2

Teacher feedback: This repeats the roster form. One valid set-builder form is B = {3^n : n is a natural number and 1 <= n <= 4}.

2. Empty set (1 mark)

Is A = {x in N : 3 < x < 4} empty?

Sample response: A is not empty.

Illustrative marks: 0 / 1

Teacher feedback: No natural number lies strictly between 3 and 4, so A is empty.

3. Finite or infinite (2 marks)

Is the set of all points on a circle's circumference finite or infinite? Give a reason.

Sample response: Finite.

Illustrative marks: 0 / 2

Teacher feedback: A circle's circumference contains infinitely many points, so the set is infinite.

4. Equal sets (2 marks)

Are A = {-2, 2} and B = {x in Z : x^2 = 4} equal? Why?

Sample response: No.

Illustrative marks: 0 / 2

Teacher feedback: Yes. The integer solutions of x^2 = 4 are -2 and 2, so B = {-2, 2} and both sets are equal.

Marked response examples (continued)

These comments are demonstration annotations, not a real teacher marking record.

5. Interval notation (2 marks)

Write (-5, 9] in set-builder form.

Sample response: {x : -5 < x < 9}

Illustrative marks: 1 / 2

Teacher feedback: The right endpoint 9 is included, so the condition must be -5 < x and x <= 9.

6. Set operations (3 marks)

If A = [-3, 5] and B = (0, 6], find A - B and A union B.

Sample response: A - B = (-3, 0] and A union B = (-3, 6)

Illustrative marks: 1 / 3

Teacher feedback: 0 is not in B, so 0 remains in A - B. The value -3 belongs to A and 6 belongs to B, so A union B includes both endpoints: [-3, 6]. Also, -3 must remain included in A - B.

7. Disjoint sets (1 mark)

Are A = {1, 2, 3, 4} and B = {x in N : 5 <= x <= 7} disjoint?

Sample response: No, they share values.

Illustrative marks: 0 / 1

Teacher feedback: They have no common elements: A contains 1 to 4 and B contains 5, 6 and 7. The sets are disjoint.

8. Cardinality (2 marks)

If n(X)=19, n(Y)=37, and n(X intersection Y)=12, find n(X union Y).

Sample response: 68

Illustrative marks: 0 / 2

Teacher feedback: Use n(X union Y) = n(X) + n(Y) - n(X intersection Y) = 19 + 37 - 12 = 44.

Total shown on the sample record	3 / 15 marks (illustrative only)

For responsible demonstrations

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